
Publication Decision from *Journal of High School Science*

"A Calibrated, Robustness-Characterized Machine-Learning Benchmark for
Threshold-Based Biosignature Labeling of Synthetic Exoplanet Transmission
Spectra"

Decision made on September 8, 2026

Editorial board's determination

Reject

Comments from the editor

Dear author,

Per the reviewer's comments, we regret that we are unable to consider your manuscript for publication in the Journal of High School Science.

Best,

Shireesh Apte

Reviewer 1

Rating scale questions

	Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
The paper makes a significant contribution to scholarship.		✓			
The literature review was thorough given the objectives and content of the article.			✓		

Open response questions

Comments to author

The revision is diligent and technically substantial, but extensive effort does not itself create novelty. After the appropriate narrowing of its claims, the manuscript establishes mainly that:

- a nonlinear classifier can recover deterministic CH_4/O_3 threshold labels from synthetic spectra;
- performance declines under noise, aerosols, stellar contamination, resolution loss, opacity changes, and other distribution shifts;
- whitening can amplify nuisance variation;
- explained PCA variance does not necessarily correspond to discriminative importance; and
- thresholds and probability calibration do not transfer reliably across domains.

These are useful demonstrations, but none is conceptually surprising. Sensitivity testing of synthetic-trained classifiers under domain shift has been performed repeatedly, and several conclusions follow directly from established machine-learning principles. The study does not introduce a new classifier, labeling framework, atmospheric model, robustness method, validation principle, or physical insight.

More importantly, the revisions eliminate much of the original scientific rationale:

- The model does not detect biosignatures.
- The labels are arbitrary deterministic thresholds rather than retrieval-derived quantities.
- The positive atmospheres are not photochemically self-consistent.
- The grid contains no habitable-zone analogues.
- The principal result is not applicable to real Ariel observations.
- The simulated resolution is substantially more favorable than Ariel's relevant channels.
- The parameter sampling is unintentionally coupled.
- The CNN comparison is not particularly meaningful because convolution is applied to PCA components, which have no wavelength locality.

Therefore, the paper may be methodologically competent but is scientifically incremental. Since JHSS requires the production/elucidation/uncovering of non-obvious new knowledge, I recommend rejection rather than another revision. Asking for retrieval-derived labels, realistic channel-level spectra, physically consistent atmospheres, and retraining under domain shift would amount to requesting a fundamentally new study.

A formal review follows:

The author has undertaken an unusually substantial and careful revision and has addressed most of the previous comments through new analyses, additional robustness experiments, appropriate qualification of the PCA and whitening interpretations, and considerable narrowing of the claims. I acknowledge the quality and seriousness of this work. However, the narrowing has also clarified that the manuscript's remaining contribution is primarily a sensitivity characterization of a classifier trained to reproduce deterministic abundance-threshold labels from a single synthetic spectral grid. Similar robustness analyses of synthetic-trained classifiers under noise, resolution changes, atmospheric nuisance factors, and domain shift have been conducted extensively, while the principal findings here—that performance deteriorates under distribution shift, that calibration and operating thresholds do not necessarily transfer, and that low-variance PCA components may retain discriminative information—are largely expected from established machine-learning principles. The study does not establish biosignature detection, abundance retrieval, applicability to Ariel observations, or a new physical or methodological principle. Because correcting the remaining central limitation would require retrieval-derived uncertain labels, physically self-consistent atmospheres, and realistically configured observational data—effectively a new study—I do not believe that another round of revision is appropriate. Despite the commendable effort and technical improvements, I recommend rejection on the basis of insufficient scientific novelty and limited knowledge contribution.
